E-Commerce EDA Report

Exploratory data analysis across customers, sessions, and orders — uncovering data quality issues, behavioral patterns, and revenue drivers.

Dataset Overview

5,050

Customers

7 columns including signup date, channel, device, age, country, premium status

30,150

Sessions

7 columns including duration, pages viewed, funnel stage, UTM source

12,100

Orders

10 columns including product category, price, discount, revenue, status

Schema Highlights

Customers

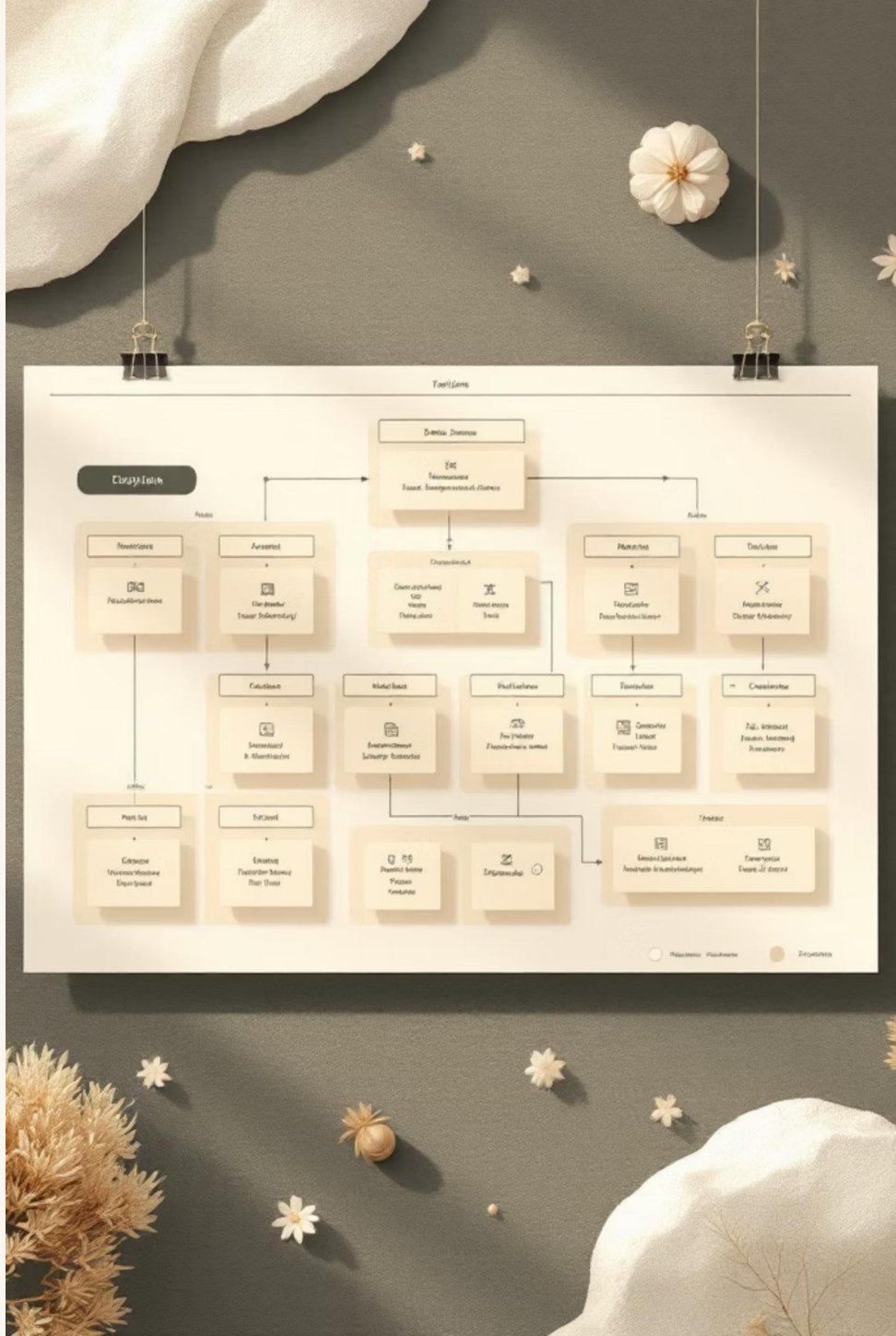
Key fields: **acquisition_channel, device_type, age_group, country, is_premium**

Sessions

Key fields: **session_duration_min, pages_viewed, funnel_stage_reached, utm_source**

Orders

Key fields: **product_category, unit_price, discount_pct, total_revenue, order_status**



Data Quality & Cleaning

Customers

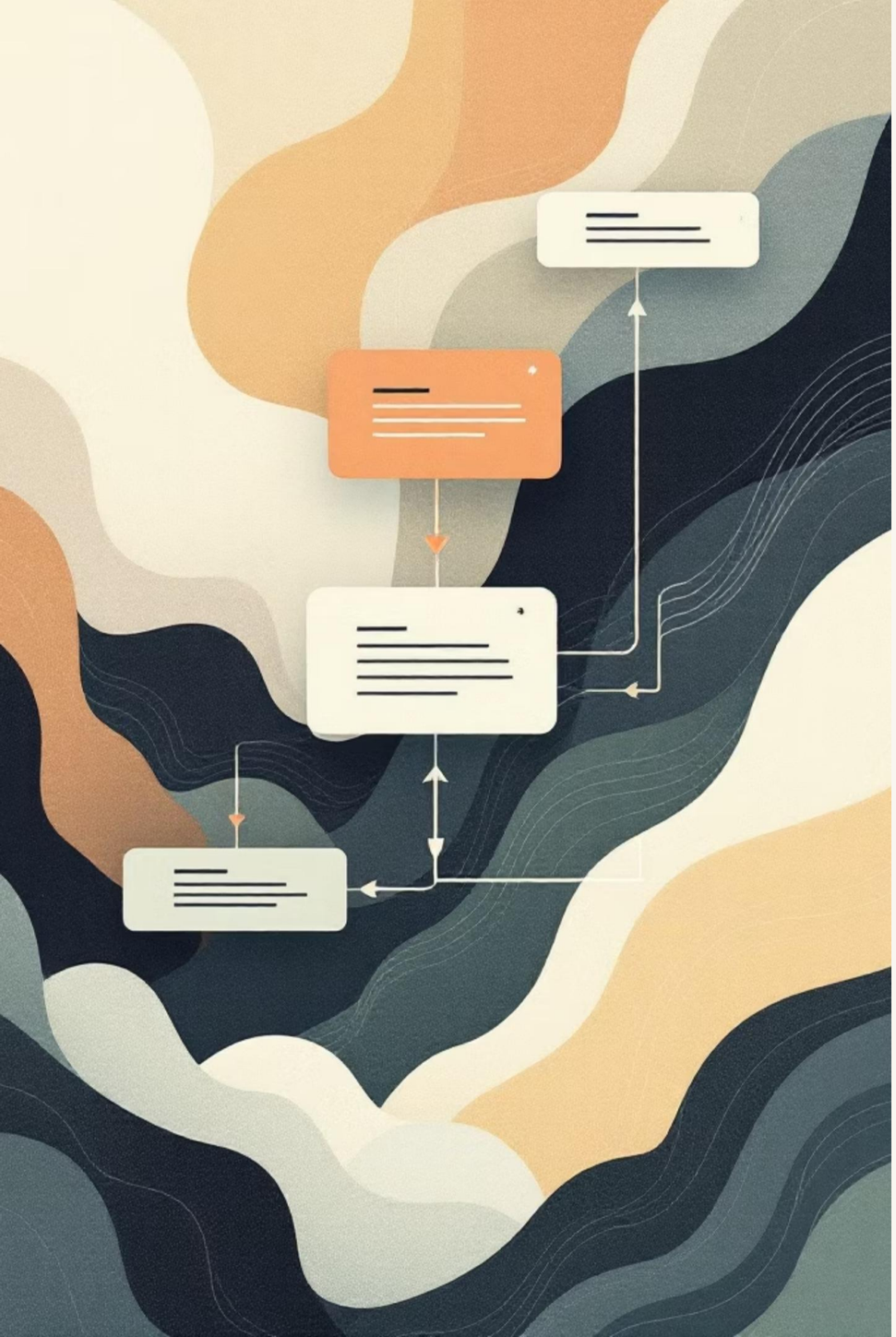
- 50 duplicate customer IDs dropped
- 251 NaN + 30 invalid "yes" values in **is_premium** corrected
- 20 invalid signup dates dropped
- ~249 NaNs each in acquisition_channel, device_type, age_group, country → filled "unknown"

Sessions

- 150 duplicate session IDs dropped
- 40 invalid session dates dropped
- 30 negative page_views dropped
- 40 negative session_duration_min dropped

Orders

- 100 duplicate order IDs dropped
- 35 negative unit prices dropped
- 25 discounts >100% dropped
- 30 invalid order_date entries dropped
- ~599 NaNs in categorical fields (product_category, payment_method, order_status) → filled "unknown"
- ~599 NaNs in numerical fields (unit_price, discount_pct) → filled with median



Data Integration & Merging

01

Orders ← Customers (Left Join)

Base table: **orders_cleaned**Join key: **customer_id**Final Merged Table: 12,068 orders retained

02

Aggregate Sessions by Customer

Source: **sessions_cleaned**Metrics: **total_sessions**, **avg_session_duration**, **avg_pages_viewed**, **funnel_stage_reached**Grain: Customer-level summary

03

Orders+Customers ← Aggregated Sessions (Left Join)

Join key: **customer_id**Missing session data: Imputed numeric fields with 0, categorical **funnel_stage** as 'no_session'
Final dataset: 11,878 complete records ready for analysis

Descriptive Statistics

After merging and cleaning, we have 11,878 complete records. Revenue distribution is severely right-skewed: the average order value is €388, but the median is only €180 — showing most customers spend far less than average. The top 1% of customers (highest-value segment) spend €2,941 or more, representing our most valuable customer tier.

Revenue

Avg €388.35 · Median €180.26 · Max €5,276.69 · 75% below €450

Engagement

Avg 6.47 sessions/customer · Median 2 · Avg duration 5.11 min · Avg 3.53 pages

Pricing & Discounts

Avg unit price €182 · Avg discount 7.83% · Deep discounts (>25%) are rare

Categorical Analysis

- Referral is the top acquisition_channel (49%), paid ads are minimal (7.5%)
- Mobile dominates (55%) but has higher return rates than desktop
- Customers are mostly 25–34 years old (31%), followed by 35–44 (22%)
- Clothing is the best-selling category (29%), followed by Electronics (23%)
- 73.5% complete · 15% returned · 6.5% cancelled
- Only 4.7% of visitors actually purchase
- Premium members are just 18% — room to grow

Distributions & Outliers

Notable Distributions

- **unit_price**: Right-skewed, peak €50–100, tail to ~€900
- **total_revenue**: Severely right-skewed; dense cloud beyond €1,500
- **avg_session_duration**: Bimodal — peaks at 0–2 min and 6–8 min
- **discount_pct**: Outliers at 25–40% reflect flash sales, not errors

⚠️ Flag: total_sessions

Median is **2**, yet isolated users show **300–550 sessions**. May indicate **bot traffic or duplicate tracking** — warrants deeper validation. Most other outliers reflect real business variation.

Bivariate Analysis

Session Duration vs. Revenue

No meaningful relationship. High-revenue orders occur at all session lengths — purchase value is driven by product choice, not browsing time.

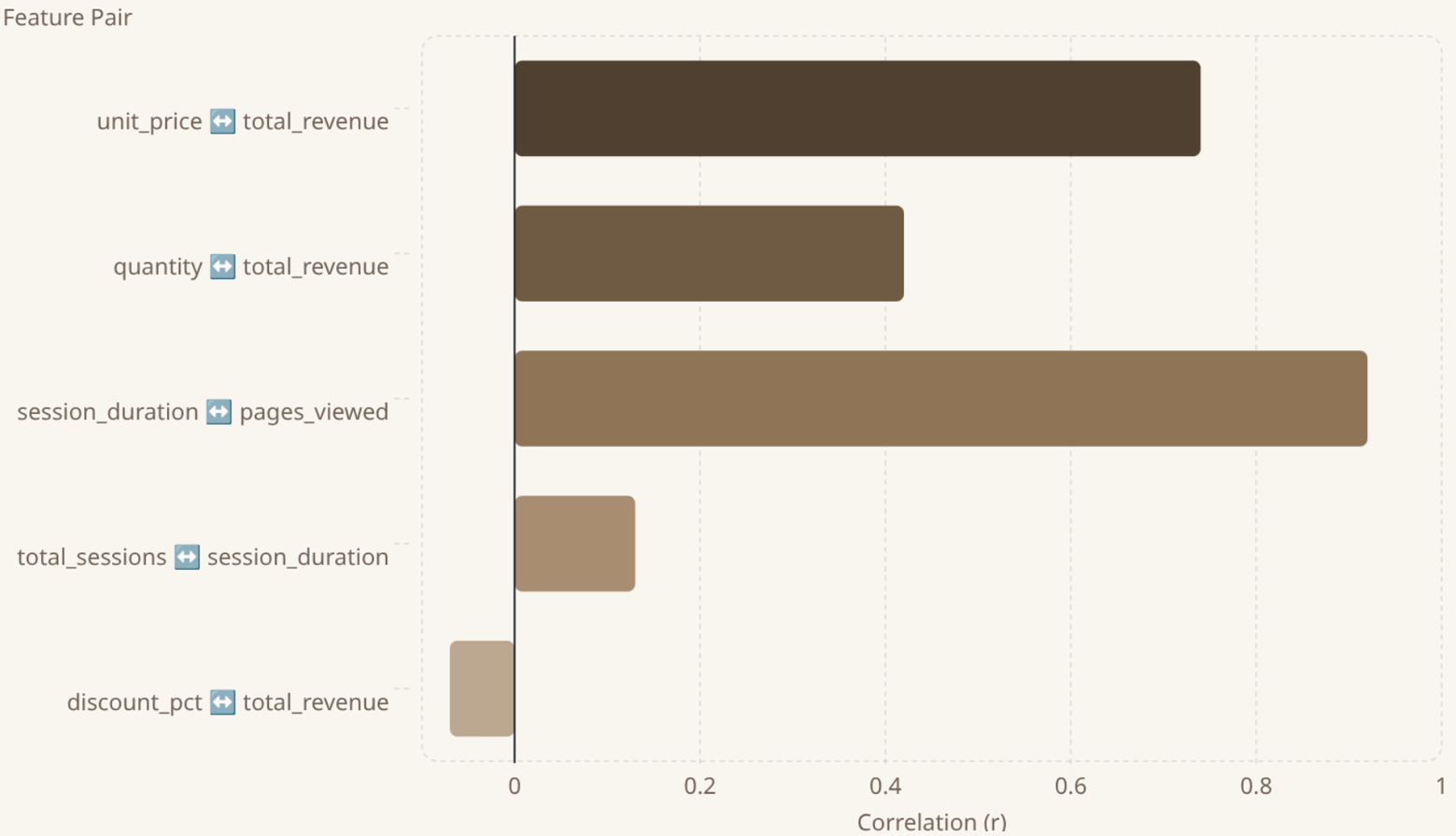
Mobile vs. Desktop Returns

Mobile leads completed order volume (4,805 orders) but has a **higher return rate ~17.8%** vs. desktop ~11.4%, indicating stronger desktop purchase intent.

Discount % vs. Revenue

Clear inverse pattern: high-revenue orders cluster at **0–5% discount**. Deep discounts (>15%) appear only on low-value transactions. Discounting is not a revenue driver.

Multivariate Analysis



Key Findings

- **unit_price ↔ total_revenue: 0.74 (Strong)** — price is the primary revenue driver
- **quantity ↔ total_revenue: 0.42 (Moderate)** — volume contributes but less than price
- **avg_session_duration ↔ avg_pages_viewed: 0.92 (Near-perfect)** — highly collinear, only use one
- **total_sessions ↔ avg_session_duration: 0.13 (Weak)** — more sessions ≠ longer sessions
- **discount_pct ↔ total_revenue: -0.07 (Negligible)** — discounts don't drive revenue

Feature Engineering

Engineered Features Overview

revenue_per_session

Heavily right-skewed; most customers generate <€200 per session, with tail to ~€4,500. Mirrors total_revenue distribution, confirming most are low-efficiency browsers.

customer_lifetime_orders

Right-skewed with most customers at 1–5 lifetime orders. Tail extends to ~140 orders. Sharp drop-off after ~10 orders confirms majority are transactional, not loyal.

day_of_week

Conversion rate flat across all days (72.6%–74.2%). <2% spread means day of week has no meaningful influence on conversion. No weekend effect.

conversion_flag vs revenue_per_session

Both converted and non-converted groups share near-identical IQR and outlier structure. Conversion flag alone is a weak signal.

Key Correlations

- **avg_session_duration ↔ avg_pages_viewed:** 0.92 (Confirmed collinearity — drop one)
- **total_sessions ↔ avg_pages_viewed:** 0.14 (Weak but strongest session-behaviour link)
- **total_sessions ↔ avg_session_duration:** 0.13 (Weak — session count ≠ session quality)
- **revenue_per_session ↔ total_sessions:** -0.13 (Slight inverse — more sessions = lower rev/session)
- **conversion_flag ↔ all features:** ~0.00 (Conversion uncorrelated with all engineered features)

Verdict & Findings

Critical Finding: conversion_flag shows zero correlation with every engineered feature — it cannot be predicted from session behaviour, discount, or lifetime order metrics alone. Conversion is driven by factors not captured in this dataset (product affinity, pricing intent).

Verdict: customer_lifetime_orders is the most informative new feature — surfaces loyal repeat-buyer segment. day_of_week adds no analytical value and can be deprioritised.

Hypothesis Testing & Key Takeaways

Hypothesis Testing Framework

H1	Does higher session duration lead to higher revenue?	Pearson correlation
H2	Do 20% of customers generate 80% of revenue (Pareto)?	Lorenz curve + Gini
H3	Do weekends have higher average order value?	Mann-Whitney U test

Results

H1 Rejected — Session Duration ≠ Revenue

Pearson r = 0.007, p = 0.443. No linear relationship between session duration and revenue.

H2 Supported — Pareto Revenue Concentration

Gini = 0.643. Revenue is concentrated in a minority of customers. **Invest in VIP retention and loyalty programs.**

H3 Rejected — No Weekend Uplift

Mann-Whitney p = 0.174. Spending is consistent across the week. Focus promotions on **category, discount level, and customer history** — not day-of-week.

Critical finding: **conversion_flag** shows zero correlation with all engineered features — conversion is likely driven by product affinity or pricing intent not captured in this dataset.